

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Development Engineer** role with **AWS Cryptography**. I bring over three years of professional experience building distributed backend systems, GenAI-powered platforms, and production-grade cloud infrastructure, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade systems that directly reflect the deep software engineering craft and security-first mindset this position demands. As a Software Engineer at Sperse, I owned the full software development lifecycle of a GenAI-powered multi-agent CRM platform — defining requirements, designing the architecture, building in Python, LangGraph, and FastAPI, and shipping with production monitoring. I leveraged GenAI (OpenAI Codex) with a custom Bitbucket MCP server to automate code review triage, eliminating 25–30 manual PR reviews per week and demonstrating exactly how GenAI can accelerate engineering transitions at scale — directly mirroring the team’s focus on leveraging GenAI to transition AWS services to post-quantum cryptography. I also engineered a FastMCP production server exposing 160+ LLM-callable backend actions with built-in security monitoring and OpenTelemetry instrumentation, built CI/CD pipelines that gate every release with safety-scoring evals, and designed multi-tiered distributed agentic workflows with durable execution across long-running cloud workloads. Prior to Sperse, as a Software Engineer at Qualitest, I built core infrastructure that required the same security focus central to AWS Cryptography’s mission. I designed a cloud-agnostic, multi-threaded Storage Service in Java and Spring Boot — handling file operations across Google Cloud Storage, S3, and MinIO via expiring signed URLs explicitly for data-in-transit protection — giving me direct, hands-on experience with secure, large-scale data transfer across cloud environments. I also orchestrated distributed large-scale jobs with AWS Step Functions for fault-tolerant error recovery across 100K+ weekly executions, and optimized PostgreSQL with window functions and materialized views to improve scalability under load. Together, these roles have prepared me to contribute meaningfully to delivering foundational cryptographic libraries integrated across AWS front-door services and protecting cloud workloads against emerging threats.

My open-source work and academic background reflect the same systems-level rigor and collaborative culture the AWS Cryptography team values. I built Dark Factory, an open-source distributed coding-agent orchestrator in Elixir and TypeScript, implementing concurrent OTP supervision trees for fault-tolerant parallel DAG execution and real-time agent status propagation across multi-machine fleets — an exercise in applying distributed systems design principles end-to-end while contributing to the open-source community. My Master of Science in Computer Science (4.0 GPA, Pace University) has given me a rigorous academic foundation in algorithms, systems design, and distributed computing that complements my production engineering experience and enables me to engage with technically deep problems — such as post-quantum cryptographic transitions — with both precision and curiosity.

I have always been passionate about **customer obsession** and **passion for invention**, and I envision myself contributing my talent to the furthering of AWS’s goal of being Earth’s most customer-centric and secure cloud. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan